COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 06

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Simplify $(p \vee (p \wedge q)) \wedge (q \vee \neg q) \vee (r \wedge F)$, using the usual precedence. Name the laws used.

(b) **[2 points]** Is $p \to r$, $q \to r$, $r \therefore p \vee q$ valid? Give a separating truth assignment if it is invalid.

(c) **[5 points]** Let u be rational and v irrational. Prove by contradiction that $u + v$ is irrational. Identify the contradictory conclusions.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** Domain: all staff. $T(x)$ means trained, $C(x, y)$ means colleagues, and $K(x, y)$ means x knows y. Translate: (i) every staff member is trained or has a trained colleague; (ii) some staff member has no two distinct colleagues who know each other. Each is worth 2 points.

(b) **[4 points]** Over a nonempty common domain, prove $\exists x \forall y (P(x) \vee Q(y)) \equiv \forall y \exists x (P(x) \vee Q(y))$. Explain why this swap works here.

(c) **[5 points]** From $\forall x (P(x) \to Q(x))$ and $\forall x (Q(x) \to \exists y R(x, y))$, derive $\forall x (P(x) \to \exists y R(x, y))$. Give named inference steps.

Problem 3: Sets and functions [12 points]

Use $\mathbb{N} = \{0, 1, 2, \ldots\}$.

(a) **[2 points]** For $f : \{0, 1, 2, 3\} \to \{a, b, c\}$ given by $f(0) = a$, $f(1) = f(2) = b$, $f(3) = a$, state the range and all preimages of b. Is f onto?

(b) **[3 points]** A function $\mathbb{N} \to \{0, 1\}$ is computable if a finite program returns its value and terminates on every input. Explain why uncomputable functions of this type must exist.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) Every function $\mathbb{N} \to \{0, 1\}$ can be computed by a finite program.

(ii) The lecture names the halting problem as an example associated with an uncomputable function.

(iii) $\mathcal{P}(\varnothing) = \varnothing$.

(iv) If $A \cap B = \varnothing$, then $A \setminus B = A$.

(v) $|(0, 1)| = |(0, 1]|$.

(vi) The range and codomain of a function must be identical.

(vii) There is an injection $\mathbb{N} \to (0, 1)$.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** Count functions $f : \{1, 2, 3, 4, 5\} \to \{A, B, C, D\}$ with $f(1) = A$ and $f(2) \neq A$.

(b) **[2 points]** For $0 \leq k \leq n$, explain by choosing and then ordering why $P(n, k) = \binom{n}{k} k!$. Include the case $k = 0$, using $0! = 1$.

(c) **[2 points]** Arrange A, B, C, D, E, F in a row with AB contiguous in that order and C, D contiguous in either order. Count the arrangements.

(d) **[2 points]** Deal unordered hands of two cards to each of three labelled players from nine distinct cards, leaving three undealt. Count the deals.

(e) **[2 points]** Count five-digit base-five numbers whose digits are nondecreasing from left to right. Digits are $0, 1, 2, 3, 4$, repetitions are allowed, and the leading digit is nonzero.

(f) **[2 points]** Count length-six nonincreasing strings over $\{0, 1, 2, 3\}$ whose first symbol is nonzero. Repetitions are allowed.

(g) **[2 points]** For a 4×2 matrix A and a 2×7 matrix B, compute only entries C_{ij} of AB with $i + j$ even, where $1 \leq i \leq 4$ and $1 \leq j \leq 7$. Each computed entry uses two scalar multiplications. Count the multiplications.

(h) **[2 points]** Count right/up unit-step lattice paths from $(0, 0)$ to $(3, 3)$ whose first step is up.

Problem 5: Combinatorial proof [10 points]

(a) **[5 points]** Use membership bit strings and a partition by subset size to prove $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ for $n \geq 0$.

(b) **[5 points]** Prove combinatorially that $\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n$.

Problem 6: Inclusion-exclusion [9 points]

(a) **[6 points]** Use inclusion–exclusion to count integers in $\{1, \ldots, 180\}$ divisible by none of $2, 3, 5$.

(b) **[3 points]** In the general inclusion–exclusion union formula, an element belongs to exactly $r \geq 1$ of the sets. Explain why its total signed contribution is one.

Problem 7: Pigeonhole principle [7 points]

[7 points] Let $a_1, \ldots, a_{2m+1}$ be a permutation of $1, \ldots, 2m+1$, where $m \geq 1$. Prove that $\prod_{i=1}^{2m+1}(a_i - i)$ is even. Show by a construction that the corresponding statement need not hold for a permutation of $1, \ldots, 2m$.

Problem 8: A small lottery [6 points]

A winning set is selected uniformly from the three-element subsets of $\{1, \ldots, 8\}$. You hold the fixed set $\{1, 2, 3\}$. Let X count your matches.

(a) **[1 points]** Describe the equally likely sample space and its size.

(b) **[1 points]** Find $P(X = 3)$.

(c) **[1 points]** Find $P(X = 0)$.

(d) **[1 points]** Find $P(X \geq 1)$.

(e) **[2 points]** Find $P(X = 1 \text{ or } X = 2)$ and explain why the two masses may be added.

Problem 9: Three sources [7 points]

A device comes from factories A, B, C with probabilities $1/2, 1/3, 1/6$. Their respective defect probabilities are $1/20, 1/10, 1/5$. These factory and defect statuses specify the experiment.

(a) **[3 points]** Find the probability of a defect.

(b) **[4 points]** Given a defect, find the factory posteriors. Which factories are most likely, and why is this not determined by defect rate alone?

Problem 10: Empty bins [10 points]

$m \geq 1$ labelled balls independently choose uniformly among $n \geq 2$ labelled bins. Let I_j indicate that bin j is empty, and let $Z = \sum_{j=1}^{n} I_j$. Write $q = ((n-1)/n)^m$.

(a) **[2 points]** Derive $P(I_j = 1)$ and $E(I_j)$.

(b) **[2 points]** Find the expected number of empty bins.

(c) **[2 points]** Find the expected number of occupied bins.

(d) **[4 points]** For distinct j, k, find $P(I_j = I_k = 1)$ and test independence. Then obtain $V(Z)$ by expanding Z^2.

Draft Paper

Mock examination working space.

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